

2. Experiment one showed synchronized nodes when using `generate_example_pow_chain.py`. However, in the second experiment, when one node stopped midway, the `generate_example_pow_chain.py` terminal only displayed one "Yay" output instead of two. Even upon restarting the stopped node, it retained its previous block records, indicating a potential flaw in the gossip protocol's handling of node failures. To address this, nodes could periodically broadcast checkpoint blocks, serving as reference points for synchronization in case of failures or disconnections.

3. In a permissionless blockchain network, dynamic peer discovery, where nodes can freely join and leave the network without a predetermined list of peers, requires specific message types. The Bitcoin P2P protocol's closest feature to this is the "getaddr" message, which nodes use to request a list of active peers from another node, ensuring nodes can identify others present in the network.